

Ex.no:6c	DATA VISUALIZATION USING POWER BI (FOOD WASTE PREDICTION)

Aim:

To clean, preprocess, and transform the **California Housing Dataset**, load the cleaned data into Power BI, and build an interactive visual dashboard to analyze housing values, income, rooms, population, and ocean-proximity patterns.

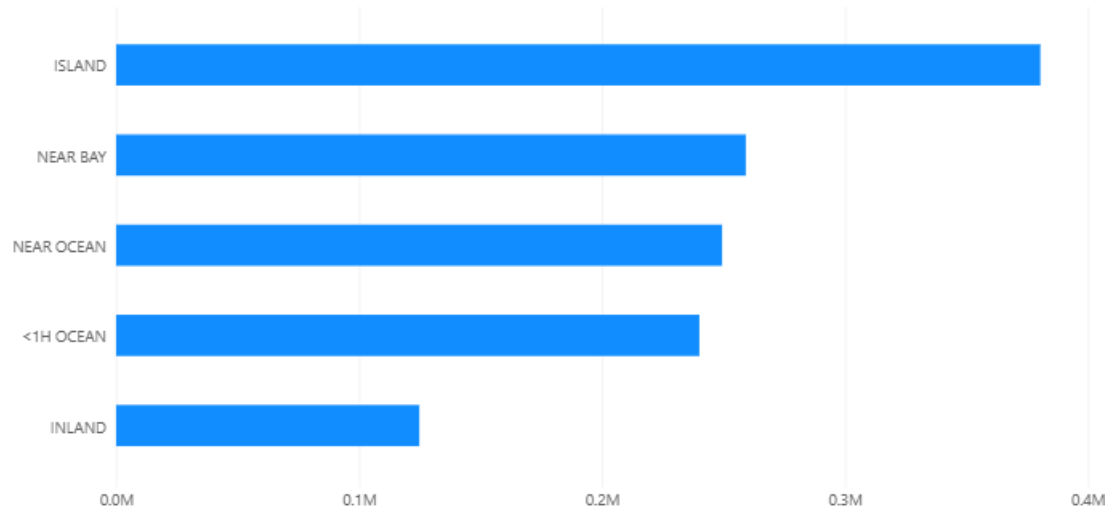
Algorithm:

1. **Import Dataset:** Load the California Housing Dataset into Power BI.
2. **Deduplication:** Identify and remove duplicate rows from the dataset.
3. **Remove Blank Rows:** Identify and remove completely blank records.
4. **Handle Missing Values:** Identify missing values and handle them appropriately to maintain data consistency.
5. **Handle Data Type Errors:** Convert numerical attributes such as total_rooms, total_bedrooms, population, households, median_income, and median_house_value into appropriate numerical data types.
6. **Clean Text Data:** Remove unnecessary spaces and standardize the ocean_proximity categorical values.
7. **Load Clean Data:** Load the cleaned California Housing Dataset into Power BI Desktop using **Get Data → Text/CSV**.
8. **Create Core Metrics:** Create measures such as total houses, average house value, minimum house value, and maximum house value.
9. **Create Dashboard:** Build interactive visualizations such as KPI cards, bar charts, column charts, donut charts, line charts, clustered column charts, and slicers to analyze housing patterns.

Output:

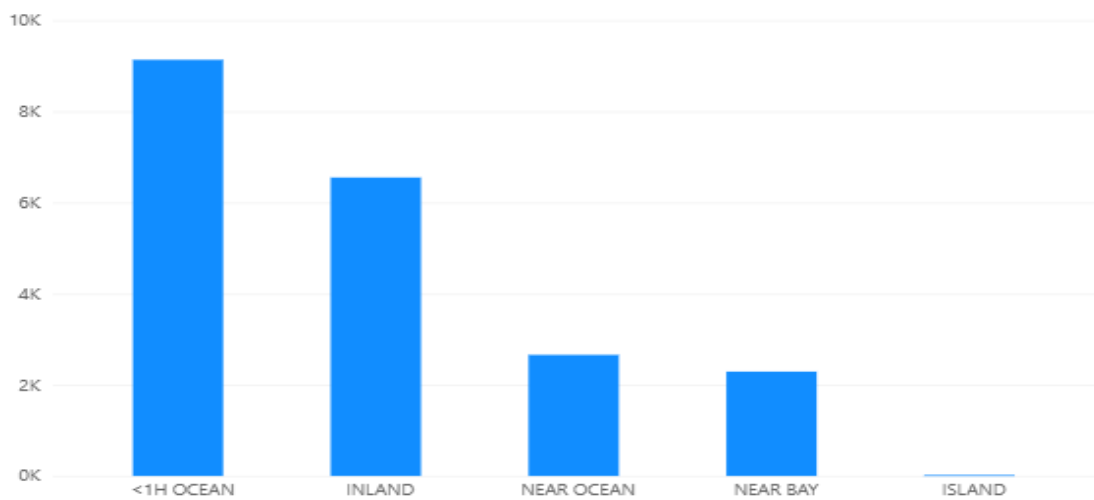
1.BAR CHART

Average of median_house_value
by ocean_proximity



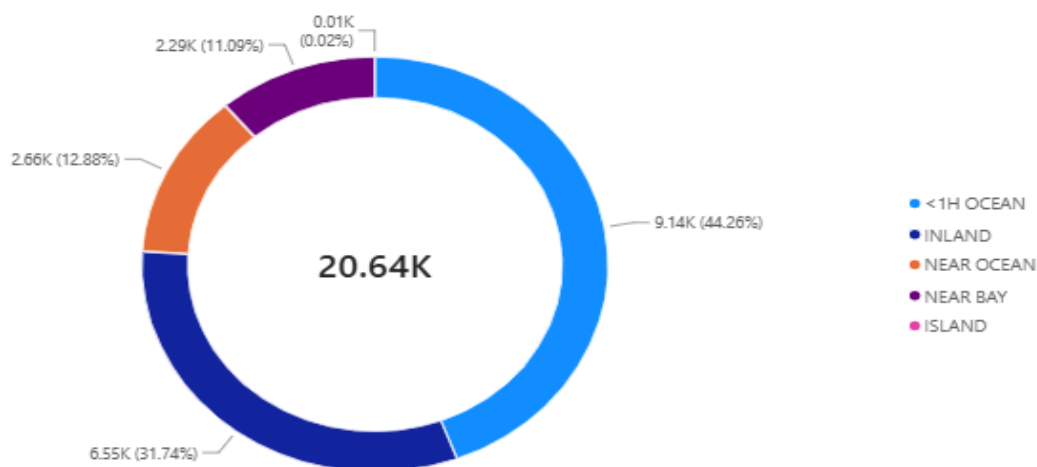
2.Column Chart

Total Houses by Ocean Proximity
Count of ocean_proximity by ocean_proximity

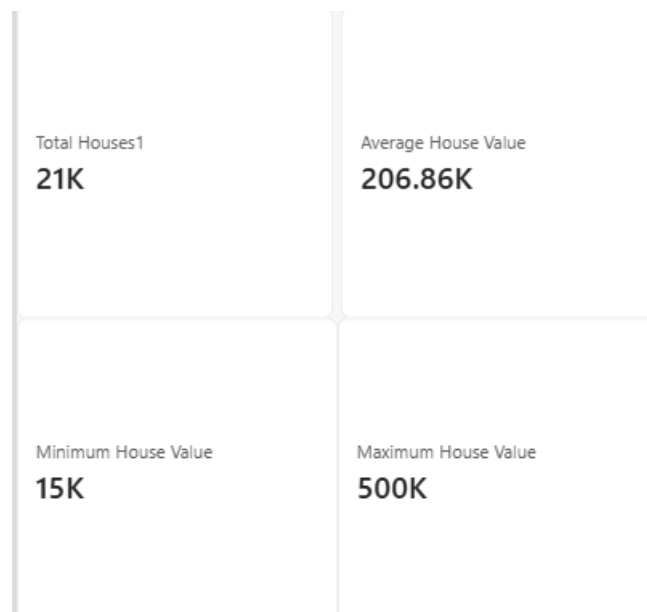


3.Donut Chart

Housing Distribution by Ocean Proximity
Count of ocean_proximity by ocean_proximity

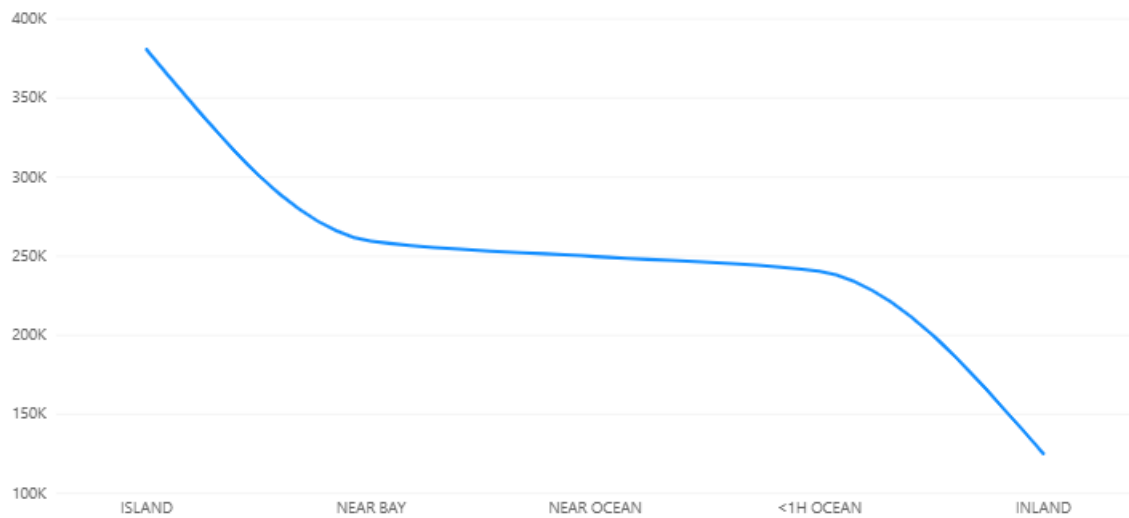


4.KPI CARD



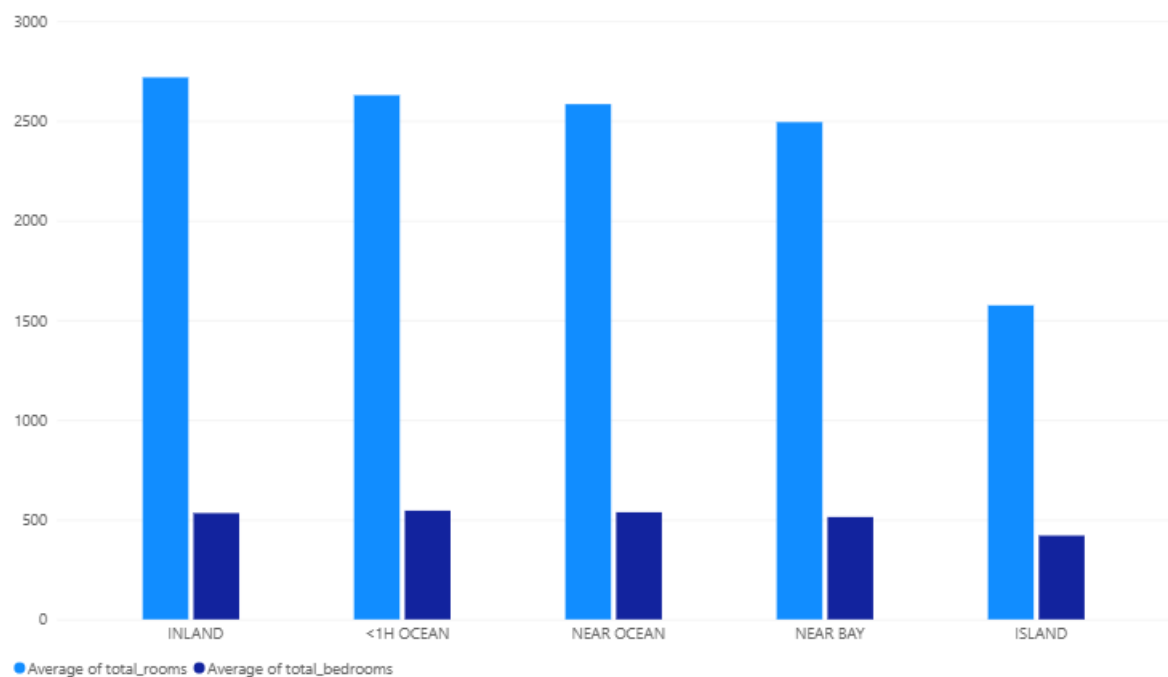
5. Line Chart

Average House Value by Ocean Proximity
Average House Value by ocean_proximity

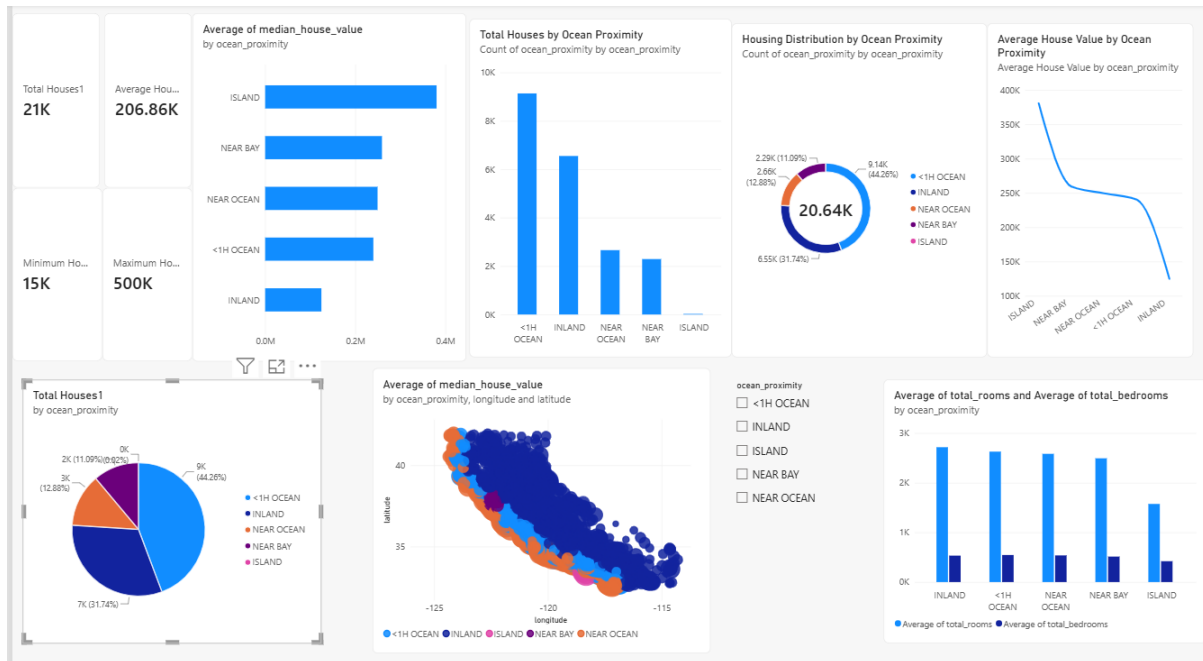


6. Clustered Column Chart

Average of total_rooms and Average of total_bedrooms
by ocean_proximity



7.DASHBOARD



Result: